

**Inventory Management and Sales Tracking System with Demand Forecasting for
Small Retail Businesses**

A Capstone Project presented to the
College of Computer Studies of
Pamantasan ng Lungsod ng Pasig

In partial fulfillment of the requirements for the degree of
Bachelor of Science in Information Technology

by:

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ABSTRACT

Title: Inventory Management and Sales Tracking System with Demand Forecasting for Small Retail Businesses

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Degree: Bachelor of Science in Information Technology

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Project Description:

This study presents the development of a web-based Inventory Management and Sales Tracking System integrated with demand forecasting to support small retail businesses in managing their operations efficiently. Many small businesses still rely on manual inventory tracking and basic record-keeping, which can result in inaccurate stock levels, overstocking, or stock shortages.

The proposed system allows business owners to monitor inventory levels, record sales transactions, and generate reports in real time. It also incorporates a demand forecasting feature using historical sales data to predict future product demand, helping businesses make informed decisions regarding stock replenishment.



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The system was developed using Agile methodology and evaluated based on ISO 25010 software quality standards, including functionality, usability, reliability, performance efficiency, security, maintainability, and portability. Evaluation results indicate that the system improves inventory accuracy, enhances decision-making, and increases overall operational efficiency for small retail businesses.



CHAPTER 1: INTRODUCTION

I. Introduction

Effective inventory management is crucial for the success of small retail businesses, as it directly impacts sales, customer satisfaction, and profitability. However, many small retailers continue to use manual methods such as handwritten logs or spreadsheets, which are prone to errors and inefficiencies.

With the advancement of information technology, automated systems can significantly improve inventory control and sales tracking. Additionally, integrating demand forecasting into these systems allows businesses to anticipate customer needs and optimize stock levels.

This study proposes an Inventory Management and Sales Tracking System with demand forecasting to provide a more efficient, accurate, and data-driven solution for small retail businesses.

II. Significance of the Study

This system will benefit the following:

- Business Owners – Better control over inventory and sales data
- Employees – Simplified transaction recording and stock monitoring
- Customers – Improved product availability
- Small Businesses – Enhanced decision-making through data insights



III. Statement of the Problem

General Problem

How can a system with demand forecasting improve inventory management and sales tracking for small retail businesses?

Specific Problems

1. How to accurately track inventory levels in real time?
2. How to record and monitor sales transactions efficiently?
3. How to predict future product demand using historical data?
4. How to generate reports to support business decision-making?

IV. Scope and Limitations

Scope

- Inventory tracking and stock management
- Sales transaction recording
- Demand forecasting using historical sales data
- Report generation and analytics
- User roles: Admin, Staff

Limitations



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- Forecasting accuracy depends on available historical data
 - Requires proper data input for reliable results
 - Limited to small retail business operations

OPERATIONAL DEFINITION OF TERMS

Administrator – The user responsible for managing system settings, inventory, and reports.

Demand Forecasting – The process of predicting future product demand based on historical sales data.

Inventory Management – The process of tracking and controlling stock levels of products.

Sales Tracking – Recording and monitoring sales transactions over time.

Database – A structured system that stores inventory, sales, and user data.

Frontend – The interface where users interact with the system.

Backend – The server-side component that processes transactions and manages data.

Stock Level – The quantity of products available in inventory.

Transaction – A recorded sale or purchase of goods.

Report Generation – The creation of summaries and analytics based on system data.

Role-Based Access Control (RBAC) – A system that limits access based on user roles.



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Performance Efficiency (ISO 25010) – The system’s ability to handle operations quickly and effectively.

Usability (ISO 25010) – The ease of use and learnability of the system.

Security (ISO 25010) – Measures to protect business data from unauthorized access.